

HANDY

Real-time ASL sign language recognition & AI-powered conversation — running fully on-device.

✕ **HOUSE LANNISTER**

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- Real-time gesture recognition
- On-device Gemma 3n LLM
- Intent detection + reply chips
- 100% offline — no cloud API

THE PROBLEM

480M people. One broken bridge.

480M**People with disabling hearing loss worldwide**

Millions communicate through sign language — but real-time, accessible tools to bridge the gap barely exist.

**Cloud Dependency**

Existing apps require an internet connection, introducing latency, privacy risk, and failure in low-connectivity areas.

**Alphabet-Only Recognition**

Most mobile apps only recognize fingerspelled letters — not full signs — making real conversation painfully slow.

**Interpreter Cost & Scarcity**

Professional interpreters are expensive and unavailable for spontaneous moments — a doctor visit, a job interview, a conversation with a stranger.

**Privacy Concerns**

Sending personal signed communication to third-party cloud servers exposes sensitive data without the user's meaningful consent.

OUR SOLUTION

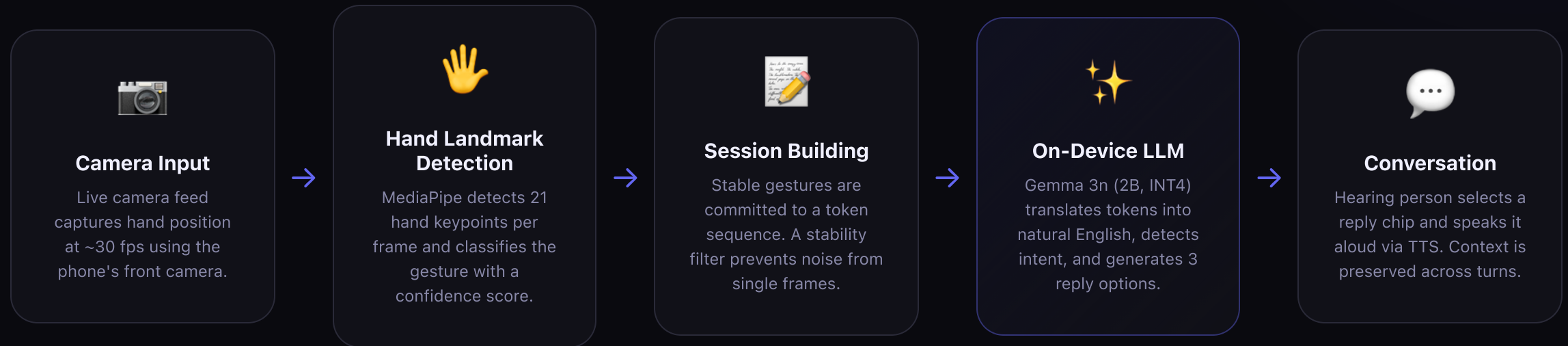
On-device ASL understanding, no internet required.

Handy is a **real-time ASL communication assistant** that runs entirely on a standard Android phone. It combines gesture recognition with an on-device large language model to translate signed gestures into natural English — and helps hearing people **respond back** in sign.

No cloud. No API key. No interpreter. **Just your phone.**

- Works fully offline — zero internet dependency
- Real-time gesture recognition at ~30 fps
- On-device Gemma 3n LLM for natural language
- Intent detection + 3 contextual reply options
- Auto-translates on signing pause (2 seconds)
- Suggests signs for the hearing person to make back

How It Works



KEY FEATURES

Built for real conversations.



Auto-Translate on Pause

Signing stops for 2 seconds → translation fires automatically. No button press needed mid-conversation.



Intent Detection Badge

Gemma classifies the emotional intent alongside translation — Question, Greeting, Request, Emotion, and more.



Three Reply Chips

Gemma generates Formal, Casual, and Empathetic reply options. Tap one → Copy or Speak it aloud via TTS.



Gesture Suggestion

Hearing person types a phrase → Gemma suggests which hand signs to make back. Full two-way communication.



Conversation Context

Previous exchanges are stored and fed back into prompts so Gemma understands the full conversational thread.



100% Offline & Private

Everything runs on-device. No data leaves the phone. Works in hospitals, schools, remote areas — anywhere.

TECHNOLOGY

Built on proven on-device AI.

LANGUAGE MODEL

Gemma 3n E2B INT4

Google's 2B parameter edge LLM, quantized to 4-bit integers via LiteRT. ~2.9 GB on device. Zero cloud calls.

ML RUNTIME

MediaPipe Tasks GenAI

Google's on-device inference runtime (v0.10.24). Handles model loading, tokenization, and streaming generation.

GESTURE RECOGNITION

MediaPipe Tasks Vision

Hand landmark detection at ~30 fps. Tracks 21 keypoints, classifies gesture with confidence score in real time.

UI FRAMEWORK

Jetpack Compose

Modern declarative Android UI with Kotlin coroutines, StateFlow, and AndroidViewModel for reactive state management.

PLATFORM

Android (Kotlin)

CameraX for live feed, TextToSpeech for audio output, CPU-based inference for broad device compatibility.

A foundation for accessible AI.

480M**People with hearing loss**

who could benefit from accessible, private, offline communication tools.

0**Cloud API calls required**

— fully offline means no latency, no cost per query, no data exposure.

2B**Model parameters on-device**

— the same intelligence as a server LLM, running in your pocket.

**Roadmap**

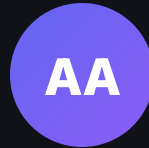
- **Custom gesture training** — expand vocabulary beyond MediaPipe defaults using a custom-trained recognizer.
- **Fine-tuned Gemma** — ASL-specific fine-tuning on gesture-to-sentence datasets for higher translation accuracy.
- **Multi-language output** — translate signed content into languages beyond English for global reach.
- **AR / wearable mode** — overlay real-time captions on smart glasses for hands-free communication.
- **Two-hand & body gesture support** — extend MediaPipe pipeline to full-body pose for richer ASL coverage.

✕ TEAM

House Lannister



Kushagra Aggarwal



Ayush Arora



Anisha Ghosh

Built with **MediaPipe** · **Gemma 3n** · **Jetpack Compose**

Fully on-device · No internet required · Android